

The background features several abstract, organic shapes in shades of purple and blue. A small circle is positioned at the top center. A large, irregular shape occupies the top right corner. Another irregular shape is located in the bottom right corner.

Introduction to Git and GitHub

Ready To Learn?

01

What will we be learning today?

What is a Version Control

Basic CLI Commands

Introduction to Git

Introduction to GitHub

Exercises

What is a Version Control System?

Let's Visualize



You are a designer



poster.png



final-poster.png



final-final-poster.png



pls-stop.png

What is Version Control System?

Version Control System manages the entire process of version and keeps the track of the changes made.

We need VCS to:

- **track the changes made,**
- **revert back to any versions when needed,**
- **compare the changes made between any two versions,**
- **to know information like who, when, why made the changes,**
- **to get all these benefits with minimal effort.**



02

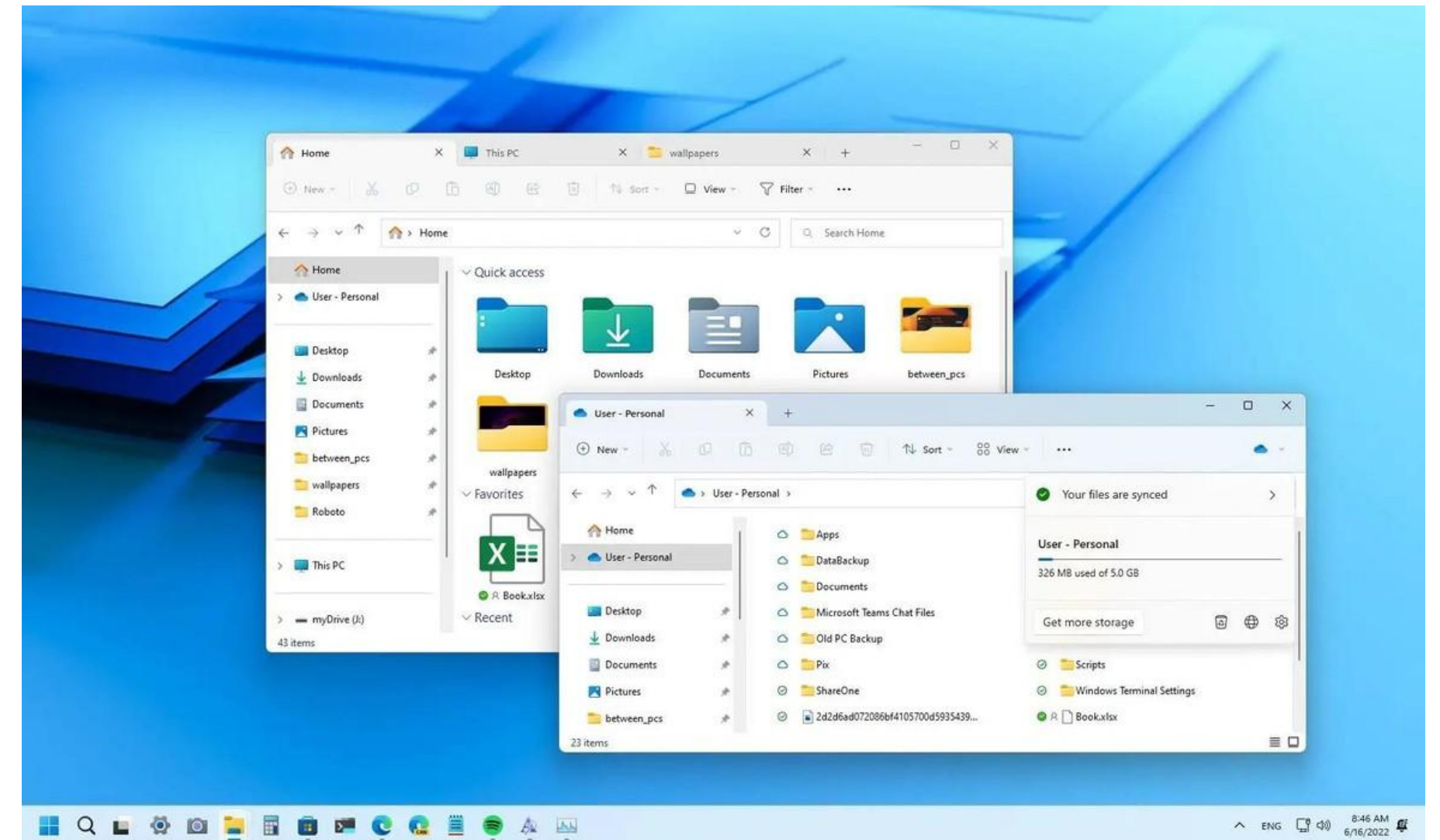
Before we learn Git
Let's learn about CLI


```
user@tecmint: ~  
File Edit View Search Terminal Help  
[user ] /home/user  
[user ] /home/user
```

Name	Size	Date	Name	Size	Date
..	4096	27 Jan 13:55	..	4096	27 Jan 13:50
..cache	4096	11 Feb 16:00	..cache	4096	11 Feb 16:00
..config	4096	11 Feb 16:36	..config	4096	11 Feb 16:36
..gnupg	4096	10 Feb 13:49	..gnupg	4096	10 Feb 13:49
..local	4096	27 Jan 13:55	..local	4096	27 Jan 13:55
..mozilla	4096	11 Feb 15:57	..mozilla	4096	11 Feb 15:57
..ssh	4096	10 Feb 13:49	..ssh	4096	10 Feb 13:49
..vim	4096	11 Feb 16:27	..vim	4096	11 Feb 16:27
Desktop	4096	27 Jan 13:55	Desktop	4096	27 Jan 13:55
Documents	4096	27 Jan 13:55	Documents	4096	27 Jan 13:55
Downloads	4096	27 Jan 13:55	Downloads	4096	27 Jan 13:55
Music	4096	27 Jan 13:55	Music	4096	27 Jan 13:55
Pictures	4096	27 Jan 13:55	Pictures	4096	27 Jan 13:55
Public	4096	27 Jan 13:55	Public	4096	27 Jan 13:55
Templates	4096	27 Jan 13:55	Templates	4096	27 Jan 13:55
Videos	4096	27 Jan 13:55	Videos	4096	27 Jan 13:55
.bash_history	57	10 Feb 17:39	.bash_history	57	10 Feb 17:39
.bash_logout	220	27 Jan 13:50	.bash_logout	220	27 Jan 13:50
.bashrc	3771	27 Jan 13:50	.bashrc	3771	27 Jan 13:50

F1=Help

Command line Interface



Graphical User Interface

Basic CLI Commands

```
$ pwd
```

Prints the full name (the full path) of current/working directory

```
$ cd Desktop
```

Changes the working directory to Desktop. Use this to navigate around CLI

```
$ ls
```

List directory contents.

```
$ mkdir name
```

Create a new directory called *name*.

```
$ rm file.txt
```

Removes the file *file.txt*

Let's Try it!



What is Git?

Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

↓ Let's Install Git

<https://git-scm.com/downloads>

Setup Git

Check version

```
$ git --version
```

Add in your name

```
$ git config --global user.name "[firstname lastname]"
```

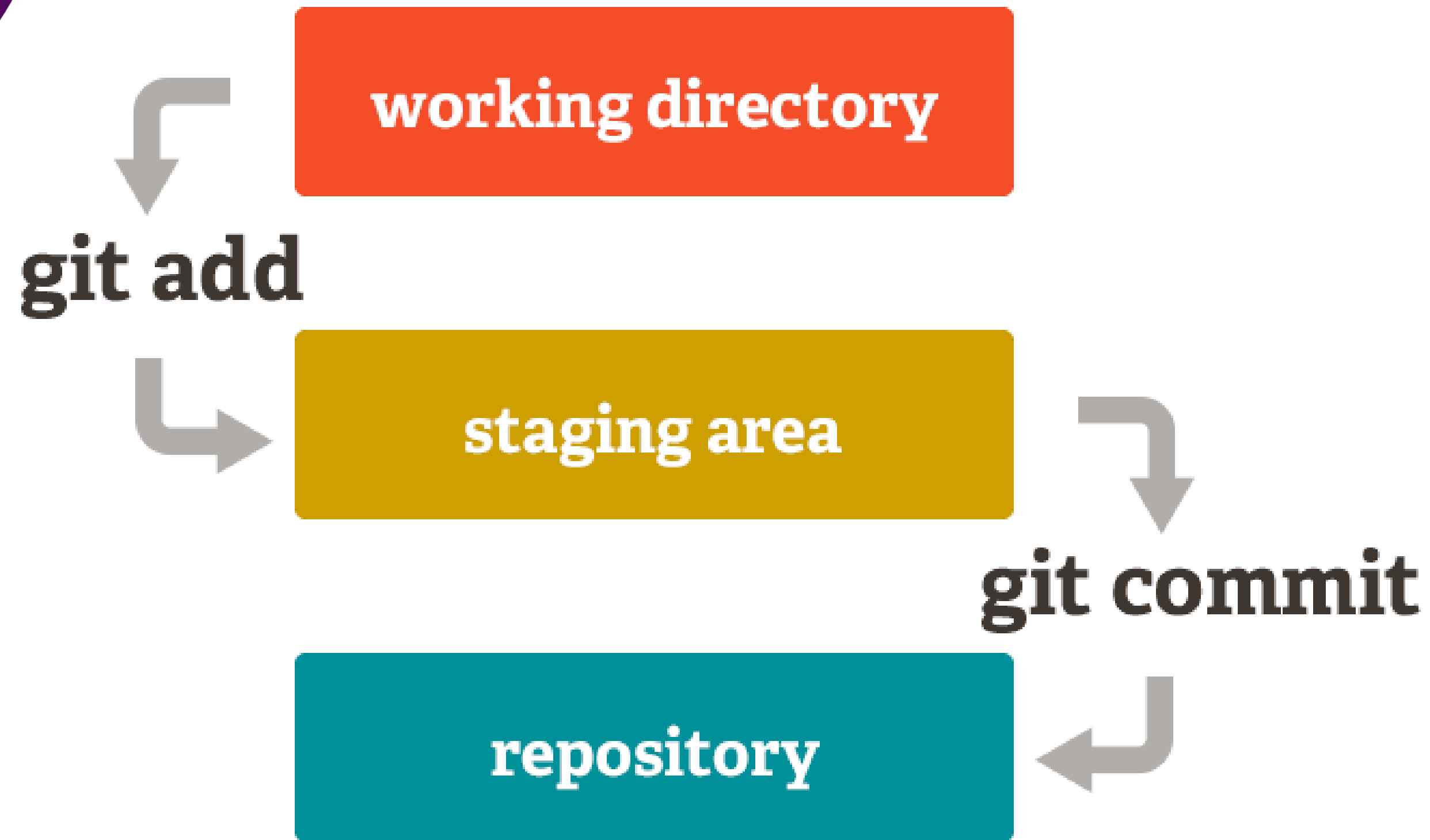
Add in your email id

```
$ git config --global user.email "[valid-email]"
```

Initialize an existing directory as a Git repo

```
$ git init
```

How Git works



Git Commands

It displays the state of working tree and staging area.

```
$ git status
```

It starts to track a file by adding it to the staging area

```
$ git add [file]
```

```
$ git add .
```

It unstages the file

```
$ git reset [file]
```

Git Commands

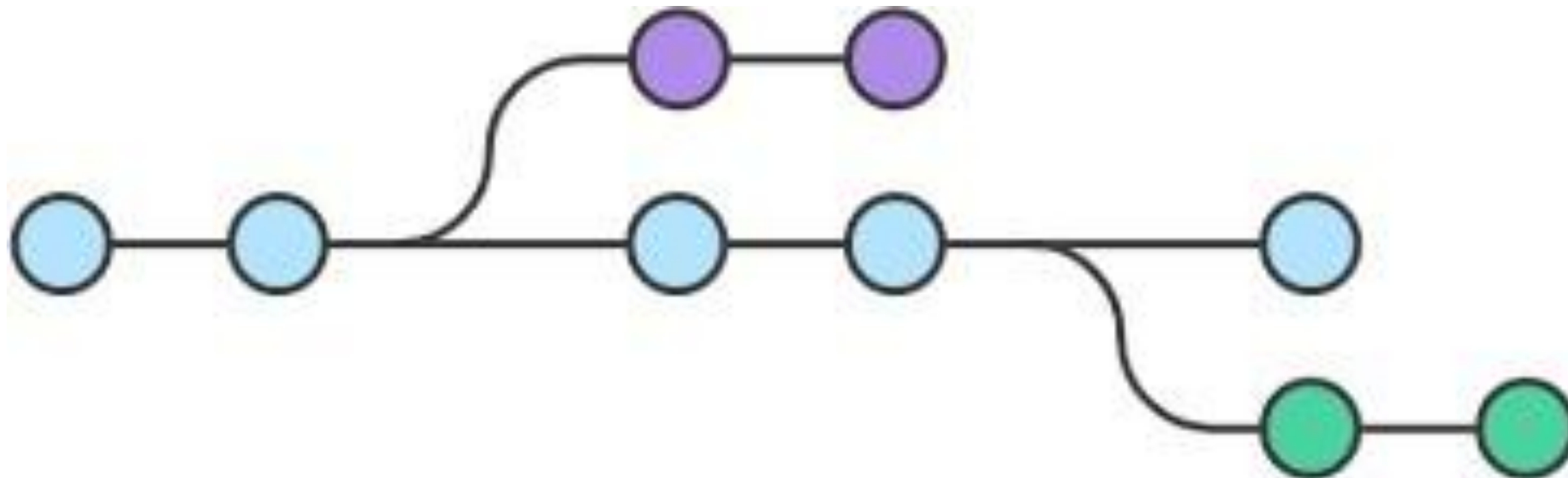
It commits your changes and captures a new snapshot

```
$ git commit -m "Some valuable message"
```

It displays all the commits in the **current branch.**

```
$ git log
```

Branching



Git Commands

List your branches

```
$ git branch
```

Creates a new branch from the current commit

```
$ git branch [name]
```

Switchs to another branch

```
$ git checkout [name]
```

Merges the specified branch's history into the current one

```
$ git merge [branch]
```

Git ignore



```
1  # .gitignore file is used to ignore some files in a repo
2  # such files always remain untracked
3  # the use of such file is done as follows
4
5  # any line starting with # is a comment
6  # ignoring file named password
7  password
8  # ignoring all files ending with .exe
9  *.exe
10 # but not ignoring final.exe
11 !final.exe
12 # ignoring all files inside build folder
13 build/
14 # ignoring file TODO not inside any subdirectory
15 /TODO
```

What is GitHub



- Till now, we have all the changes made saved in our local system only.
- To host our repository online, we need a server or hosting platform.
- GitHub is a repository hosting platform that uses Git in its core.

Let's make an account!

Some More Git Commands

Add a remote repo link

```
$ git remote add origin [link]
```

Push local repo changes to remote repo

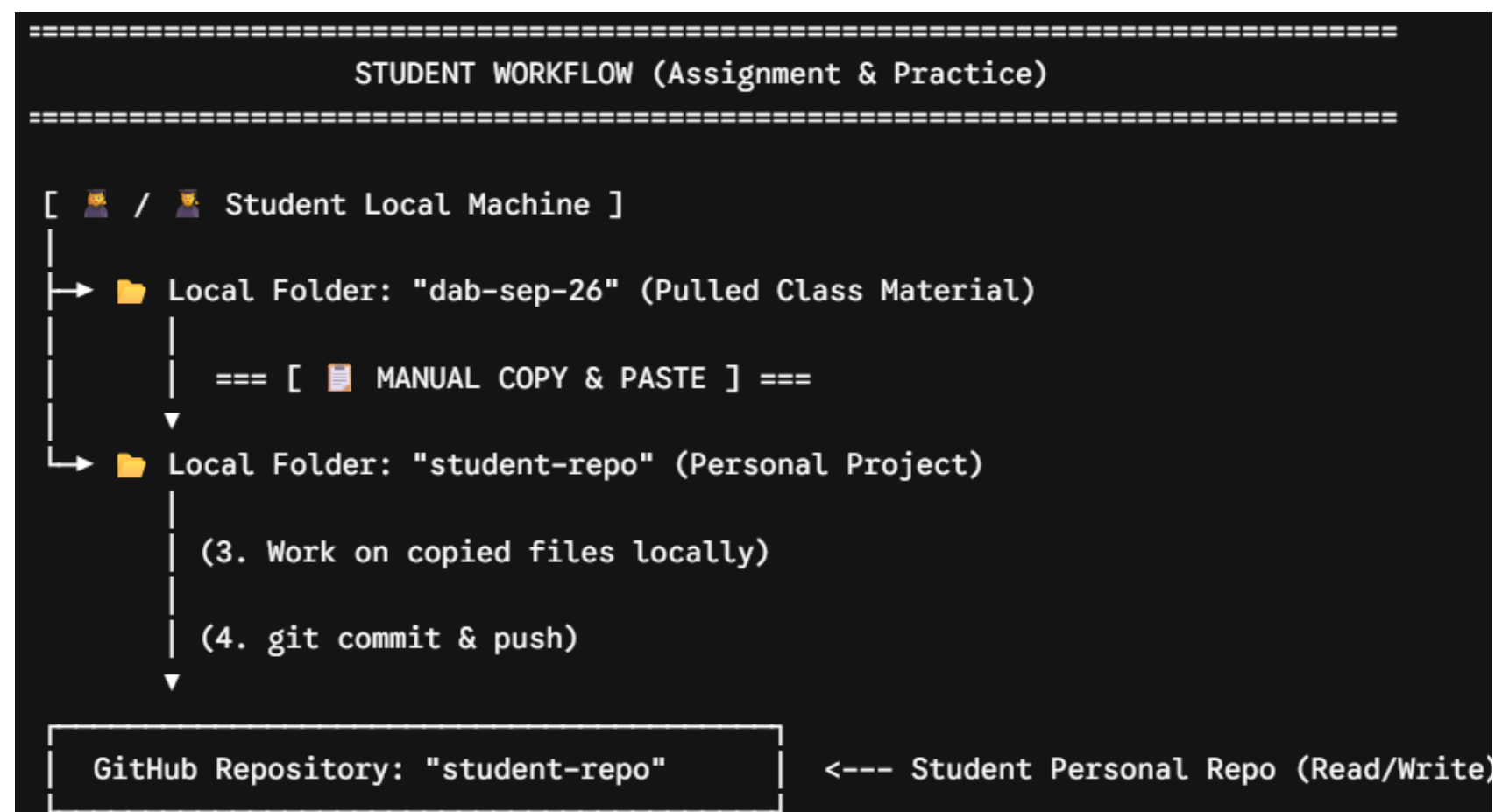
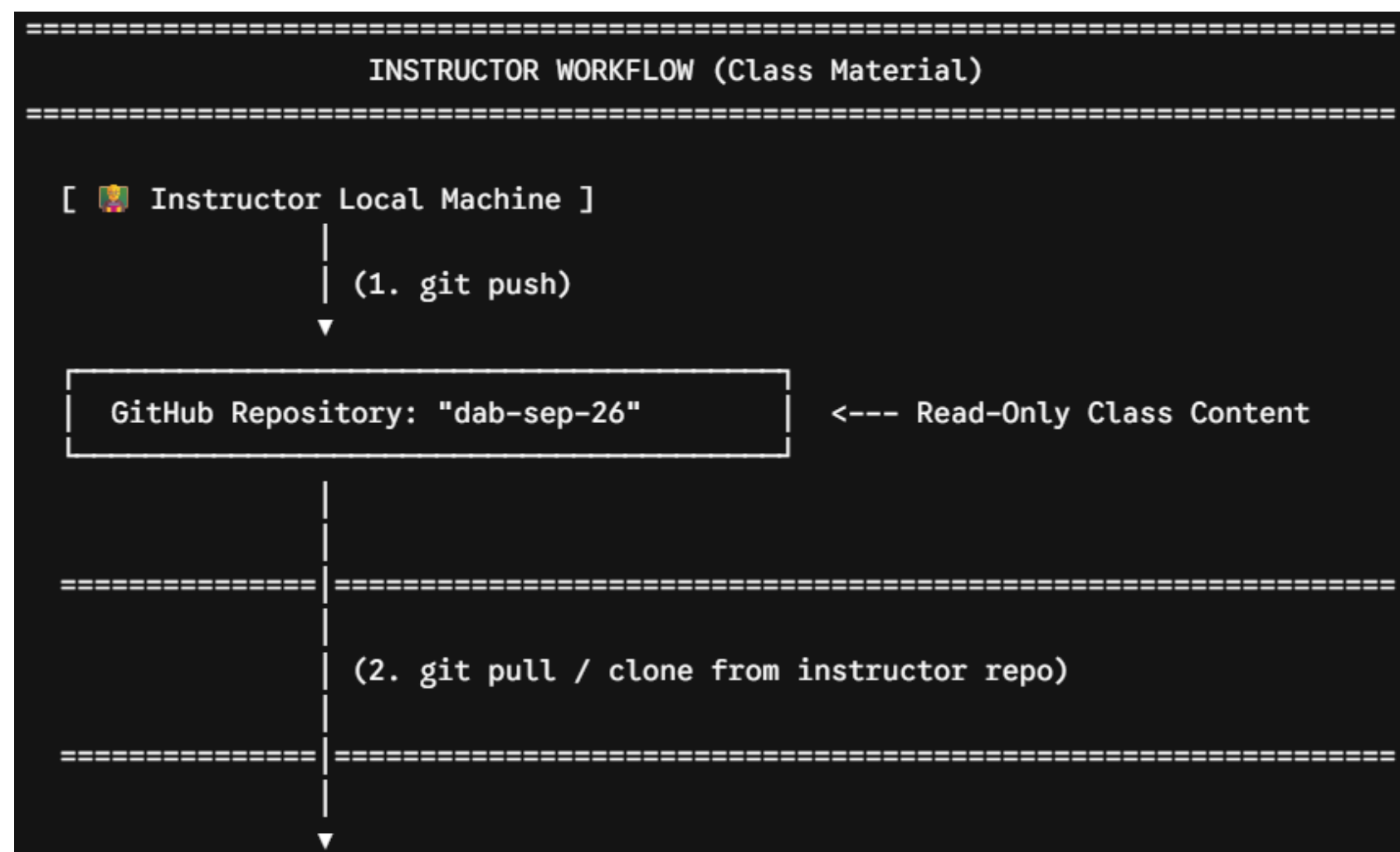
```
$ git push
```

Fetch and merge changes from remote repo

```
$ git pull
```

Retrieve an hosted repository from a URL

```
$ git clone [link]
```



Key Workflow Sequence:

1. **Instructor:** Pushes class notebooks and materials to `dab-sep-26`.
2. **Student Pull:** Students run `git pull` (or `git clone`) to sync the latest content from `dab-sep-26` to a local read-only reference folder.
3. **Local Manual Transfer:** Students copy the required files from their local `dab-sep-26` folder into their local `student-repo` working directory.
4. **Student Push:** Students work on those files and execute `git push` to upload their completed work to their personal GitHub repository.

```
$ git add [file]
$ git add .
```

```
$ git commit -m "Some valuable message"
```

```
$ git push
```




Thank You!